Theorem 1. Suppose $\mathfrak{g}$ is a semisimple lie algebra, then for any $\mathfrak{g}$-module V we find that V is completely reducible.

Proof. I will follow the proof in Humphrey's, this uses the clever observation that the case of codimension 1 is sufficient. Assume the result in codimension 1, then let $0 \neq W \subsetneq V$ be a submodule. $\hom(V, W)$ is a $\mathfrak{g}$-module with submodules

$$\Lambda := \{T \in \hom(V,W) \mid T|_W = \lambda \in k\} \quad \Lambda_0 := \{T \in \hom(V,W) \mid T|_W = 0\} \tag{1}$$

to see these are submodules let $\xi \in \mathfrak{g}$, $T \in \hom(V, W)$ and $w \in W$

$$(\xi.T)(w) \overset{\text{def}}{=} \xi.T(w) - T(\xi.w) = \lambda T(w) - T(\lambda w) = 0 \tag{2}$$

Moreover it is immediate that $\Lambda/\Lambda_0 \cong k$. Now the result in codimension 1 gives a decomposition of Λ into $\mathfrak{g}$-submodules $\Lambda_0 \oplus kT$ with T chosen so that $T|_W = 1$. It follows that $V = \ker T \oplus W$, to conclude this gives reducibility we check that $\ker T$ is a submodule, so let $v \in \ker T$ and $\xi \in \mathfrak{g}$, then

$$T(\xi.v) \overset{\text{defn of hom}}{=} \xi.T(v) = \xi.0 = 0 \tag{3}$$

so that $\xi.v \in \ker T$.

To show the result holds for a codimension 1 submodule, we will first prove it in the case that W is irreducible: Let c_ϕ be the Casimir element of ϕ, then c_ϕ commutes with $\phi(\mathfrak{g})$. Moreover, since V/W is one dimensional and $\mathfrak{g}$ is semisimple, we have that $\phi(\mathfrak{g}) = \phi([\mathfrak{g},\mathfrak{g}]) = [\phi(\mathfrak{g}), \phi(\mathfrak{g})] \subset \mathfrak{sl}(V)$ but in dimension 1 we have $\mathfrak{sl}(V) = 0$ so that $\phi(\xi) : V \to W$ for all $\xi \in \mathfrak{g}$. Since c_ϕ is a sum of compositions of elements of $\phi(\mathfrak{g})$ we conclude that $c_\phi : V \to W$. Since W is irreducible by Schur's lemma the Casimir element acts as a scalar on W it follows the casimir element is of the form

$$\begin{pmatrix} 0 & \mathbf{0} \\ * & \frac{\dim \mathfrak{g}}{\dim W} 1_{(n-1)\times(n-1)} \end{pmatrix} \tag{4}$$

thus has rank $n-1$. Moreover $\ker c_\phi$ is a $\mathfrak{g}$ submodule since c_ϕ commutes with $\phi(\mathfrak{g})$ It follows that $V = W \oplus \ker c_\phi$ is a decomposition.

Now we proceed to prove the result in the case of V having a codimension 1, but not necessarily reducible submodule W, to do so we prove the following statement by induction:

"Every (n-dimensional) $\mathfrak{g}$ module V with a codimension 1 submodule W is completely reducible"

The case $n = 2$ is trivial. Now by induction suppose that V is not irreducible, and has a submodule W of codimension 1. Then if W is irreducible we are done, if not W has an irreducible submodule W', it follows that W/W' has a codimension 1 submodule in V/W', which we denote $\hat{W}/W'$, since $\dim \hat{W} < \dim V$ we know that $\hat{W} = W' \oplus X$ where X has dimension 1. Thus we get the decomposition

$$V = W \oplus X \tag{5}$$

and W is decomposable by induction. □